

Sujith S

Applied Scientist | Agentic AI · NLP · Multimodal ML · Generative AI · Deep Learning
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Summary

Published researcher and Computational Statistics student at VIT University (2022-2027) with applied experience building agentic AI systems, NLP pipelines, multimodal deep learning architectures, and Retrieval-Augmented Generation (RAG) systems. Proficient in Python, PyTorch, TensorFlow, and Hugging Face Transformers; experienced designing multi-step reasoning agents, autonomous task execution workflows, and representation learning pipelines for production environments on AWS. Passionate about building Generative AI solutions that solve complex business problems at scale, with a focus on trust, safety, and responsible AI.

Education

Int.MSc Computational Statistics and Data Analytics

VIT University | 2022-2027

Coursework: Machine Learning, NLP, Deep Learning, Statistical Modeling, Data Structures & Algorithms, Distributed Systems

CS50 Programs

Harvard CS50 (Online)

Completed [CS50x](#), [CS50P](#), [CS50AI](#), [CS50SQL](#), and [CS50R](#) covering algorithms, AI, Python, and full-stack systems.

Skills

Languages: Python, Java, SQL, R, JavaScript, TypeScript

Machine Learning & AI: Agentic AI, Multi-Step Reasoning, Deep Learning, Reinforcement Learning, NLP, NLU, LLMs, Multimodal Learning, Representation Learning, Few-Shot Adaptation, SFT, Model Compression, Inference Optimization

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, CrewAI, AutoGen, FAISS, ChromaDB, Scikit-learn, XGBoost, Sentence Transformers, NLTK

Data & Infrastructure: Apache Kafka, Apache Spark, Delta Lake, PostgreSQL, Redis, MLflow, Docker, AWS (Lambda, S3, SageMaker, EKS), CI/CD Pipelines, Linux/Unix

Research & Publications

Stock Market Analysis and Prediction Using Machine Learning

2025

[El Ruha International Conference on Scientific Research](#), pp. 730-740. Built a unified ML framework applying LSTM, GRU, Transformer, ARIMA, XGBoost, LightGBM, and Reinforcement Learning agents (DQN, PPO, A2C) for financial time-series forecasting; designed representation learning pipelines and evaluated 21 models across diverse architectures on real market data. [\[GitHub\]](#)

Projects

Multi-Agent Business Automation System

[GitHub](#)

Python, CrewAI, LangGraph, LangChain, AutoGen, FastAPI, PostgreSQL, Redis, Docker, AWS, Prometheus

- Built an enterprise agentic AI platform where specialized autonomous agents (Finance, Analytics, Operations, Support) collaborate under a Supervisor Agent for multi-step reasoning, task delegation, and policy optimization; implemented reinforcement learning techniques for agent decision-making, contextual memory, intelligent task routing, and failure recovery across distributed agentic workflows.

Autonomous Research Agent (RAG + Agentic NLP)

[GitHub](#)

Python, LangChain, LangGraph, Hugging Face Transformers, Sentence Transformers, FAISS, ChromaDB, FastAPI, PostgreSQL, Docker, AWS, AutoGen

- Architected an agentic RAG system with multi-step NLU reasoning using LangChain, LangGraph, and AutoGen; implemented self-supervised representation learning via Hugging Face Sentence Transformers for semantic chunking, vector embedding generation, and cross-modal similarity search via FAISS and ChromaDB; automated evaluation pipelines benchmarking retrieval precision and few-shot adaptation performance.

Social Media Sentiment Lakehouse

[GitHub](#)

Python, Hugging Face Transformers, Apache Kafka, Spark Streaming, Delta Lake, NLTK, MLflow, PostgreSQL, Docker, AWS

- Designed a large-scale multimodal NLP analytics platform ingesting real-time unstructured text streams via Apache Kafka; implemented Hugging Face Transformer pipelines for sentiment classification, emotion detection, and topic modeling; built end-to-end NLU preprocessing and inference modules with MLflow experiment tracking and production deployment on AWS.

Work Experience

Founder

[Shoot the Breeze](#) |
2023 - Present

- Owned end-to-end agentic ML system development: designed NLP-powered autonomous features, implemented secure and testable model training and inference pipelines in Python, and deployed production services on AWS with Docker, CI/CD automation, and monitoring workflows.

App/Web Developer

[CropSky](#) | VIT-TBI Research & Innovation Hub |
2024 - Present

- Developed ML-powered backend systems and REST APIs on cloud infrastructure at a university-incubated startup; implemented automated multimodal data pipelines and collaborated cross-functionally to integrate model outputs into production applications.

Certifications & Achievements

- [Google Generative AI Leader](#)
- [Google Cloud Computing Foundations](#)
- [FreeCodeCamp Legacy Full Stack](#)
- [OCI 2025 Certified Developer Professional](#)
- AWS [Solutions Architect](#) and [Developer](#)
- [Software Engineer Certification - HackerRank](#)